Endogenously sticky price

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Abstract

In this paper, we construct a monetary model with manufacturing and service sector. First it is shown that the price stickiness can be expressed in terms of previous period variables i.e. the price stickiness is endogenous. We then derive the log linearized equations characterizing a model with manufacturing and service sector. The manufacturing sector relies on inventory storage for delivering goods, whereas the service sector goods can be delivered directly to the consumer. In model manufacturing sector employment and service sector employment response to the monetary policy shock are of opposite sign. The relative strength of response depends upon the elasticity of substitution between manufacturing and service goods.

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1 Introduction

The seminal work by Gali and Monacelli (2005) has been the foundation for monetary policy analysis in much of the recent research. A key assumption in this framework is exogenously sticky price. New Keyneysian models greatly emphasize the importance of price stickiness for monetary policy transmission. It is also worth noting that observed nominal rigidity does not necessarily imply that nominal shocks have real effect, see for example, Caplin and Spulber (1987) and Golosov and Lucas (2007).

Since then a number of studies have attempted to study the importance of price stickiness in monetary policy transmission. A recent empirical analysis by Singh, Suda, and Zervou (2022) revealed the different sectors respond differently to monetary policy.

In this paper, I explore the theoretical reason behind the observed heterogeneous response. The hypothesis is that different sectors have different price stickiness that arises from inventory and shelf life of the product.

Relevance to Emerging Economies: The main motivation for this paper is derived from empirical work based on the US dataset. However the framework will be useful for policy research in Emerging Economies where agriculture is a dominant sector. For example, Ghate et al. (2018) and Bahl et al. (2025) build a multisector model with agriculture and manufacturing. The agriculture sector is assumed to have flexible price while manufacturing sector prices are sticky. This assumption is similar to the inventory model of this paper where we model service sector as having flexible price while prices of the manufatured goods are sticky.

Relation to the literature

This work builds upon the existing literature on inventory management. Some of the early works in this area were done in Arrow et al. (1951), and Bellman et al. (1955). They derive optimal inventory policy under various conditions, however, assuming demand function to be exogenously given. Scarf (1960), and Iglehart (1963) explore the optimality of (s,S) inventory policy in a dynamic set up.

The (s,S) policy of inventory management is one of the widely studied model by economists. Iglehart (1963) study convergence of economies aggregate to frictionless counterpart. Khan and Thomas (2007) incorporate (s,S) into a business cycle model to explain cyclical variability of inventory investment.

The inventory management problem has also been extensively explored in operations research literature. This paper is close to continuous time models as described in Benkherouf and Gilding (2021a), Benkherouf and Gilding (2021b), and Alfares and Ghaithan (2016). The contribution of this paper is to study the inventory decision with non linear price dependent demand rate.

Rest of the paper is organized as follows. In Section 2, I present the benchmark New Keynesian model for reference and comparison. Section 3, I modify the benchmark model by eliminating the exogenous price stickiness, then introduce two sectors, manufacturing and service. The manufacturing sector has manufacturing, for production, and retail unit, for inventory accumulation to fulfill future demand.

2 A model with sticky price

Household

There are a continuum of of mass 1 household that maximizes their lifetime utility over consumption and leisure. The expected lifetime utility is given by:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t), \tag{1}$$

where C_t is a consumption index given by

$$C_t = \left(\int_0^1 C_t(i)^{1-\frac{1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}}. \tag{2}$$

I assume the following functional form for $U(C_t, N_t)$,

$$U(C_t, N_t) = \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \psi \frac{N_t^{1+\phi}}{1+\phi} \right), \tag{3}$$

and β is the discount factor, $\sigma > 0$ is that elasticity of intertemporal substitution, and $\phi > 0$ is the Frisch elasticity of labor supply. The household's optimization problem is subject to the budget constraint given by:

$$P_t C_t + Q_t B_t = W_t N_t + B_{t-1} + P_t \mathcal{F}_t \tag{4}$$

where B_t is the number of shares of government bond, Q_t is the price of government bond, and $\mathcal{F}_t$ is the profits of the firm.

Let $\Pi_t = \dfrac{P_t}{P_{t-1}}$ denote the inflation rate. The households' choice can be characterized by the following first order conditions

$$-\frac{U_{n,t}}{U_{c,t}} = \frac{W_t}{P_t} = \psi C_t^{\sigma} N_t^{\phi}, \tag{5}$$

$$Q_t = \beta \mathbb{E}_t \left(\frac{U_{c,t+1}}{U_{c,t}} \frac{P_t}{P_{t+1}} \right). \tag{6}$$

The final good producer

The final good (Y_t) is produced using the intermediate goods Y_{it} using the following technology:

$$Y_t = \left(\int_0^1 Y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{7}$$

where ε is the elasticity of substitution.

Final good producers are perfectly competitive in the input and output market. They maximize profits subject to the production function (7). Therefore the input demand function is given by:

$$Y_{it} = \left(\frac{P_{it}}{P_t} \right)^{-\varepsilon} Y_t \quad \forall i \tag{8}$$

and the price index is

$$P_t = \left(\int_0^1 P_{it}^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}}. \tag{9}$$

Intermediate good producers

Each intermediate firm produces a differentiated good using only labor as input and a linear production technology:

$$Y_{it} = A_t N_{it},$$

where ℓ_{it} is the labor hired by the firm i at time t. The productivity $a_t = \log(A_t)$ follows the following AR(1) process:

$$a_{t+1} = \rho a_t + \epsilon_{at}. \tag{10}$$

The intermediate good producers are monopolistically competitive in output market, and they reoptimize their prices P_{it} at time t with probability $1 - \theta$, with $0 < \theta < 1$. A firm which does not reoptimize the price continue with their old prices. A firm that reoptimizes its price at time t, maximizes the current real value of profits generated while the price remains effective:

$$\max_{X_{it}} \mathbb{E}_t \sum_{\tau=t}^{\infty} \theta^{(\tau-t)} Q_{t,\tau} \left(\frac{X_{it}}{P_\tau} Y_{i\tau|t} - MC_\tau Y_{i\tau|t} \right)$$

$$\text{s.t. } Y_{i\tau|t} = \left(\frac{X_{it}}{P_\tau} \right)^{-\varepsilon} Y_\tau,$$

where,$Q_{t,\tau}$ is the stochastic discount factor

$$Q_{t,\tau} = \beta^{\tau-t} \mathbb{E}_t \left[\left(\frac{C_\tau}{C_t} \right)^{-\sigma} \frac{P_t}{P_\tau} \right]$$

MC_τ is the real marginal cost of the intermediate good producers which is same across all firms

$$MC_t = \frac{W_t}{P_t A_t}.$$

Now, the first order condition of the firm with respect to p_{it} is

$$\mathbb{E}_t \sum_{\tau=t}^{\infty} \theta^{(\tau-t)} Q_{t,\tau}(1-\varepsilon)\left(\frac{X_{it}}{P_\tau}\right)^{-\varepsilon} \frac{Y_\tau}{P_\tau} = -\mathbb{E}_t \sum_{\tau=t}^{\infty} \theta^{(\tau-t)} Q_{t,\tau} MC_\tau \varepsilon \left(\frac{X_{it}}{P_\tau}\right)^{-\varepsilon-1} \frac{Y_\tau}{P_\tau}$$

$$\Rightarrow X_{it}\mathbb{E}_t \sum_{\tau=t}^{\infty} \theta^{(\tau-t)} Q_{t,\tau} \frac{Y_\tau}{P_\tau^{1-\varepsilon}} = \frac{\varepsilon}{\varepsilon-1}\mathbb{E}_t \sum_{\tau=t}^{\infty} \theta^{(\tau-t)} Q_{t,\tau} \frac{W_\tau}{A_\tau}\frac{Y_\tau}{P_\tau^{1-\varepsilon}}$$

New Keynesian Phillips Curve

The New Keynesian Phillips curve for this model can be derived by taking first order approximation around zero inflation steady state. Let the normalized price index in zero inflation steady state be 1. In addition, $Y_\tau = Y$, $Q_{\tau,t} = \beta^{\tau-t}$. Then the first order approximation around this steady state may be written as

$$x_{it} = (1-\beta\theta)\sum_{\tau=t}^{\infty}(\beta\theta)^{\tau-t}\mathbb{E}_t p_\tau^* \tag{11}$$

where, $p_\tau^* = \log(P_\tau^*)$, and $P_\tau^* = \dfrac{\varepsilon}{\varepsilon-1}\dfrac{W_\tau}{A_\tau}$ is the profit maximizing price of firm i in period τ.

Claim 1. *Around the flexible price equilibrium, the aggregate production function for this economy can be written as $Y_t = A_t N_t$ upto a first order approximation.*

Proof. content... ■

Claim 2. *The New Keynesian Phillips curve for the model with sticky price is given by*

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \pi_{t+1}, \tag{12}$$

where $\kappa = \dfrac{(1-\theta)(1-\beta\theta)(\sigma+\phi)}{\theta}$, and $\tilde{y}_t$ denotes the output gap i.e. deviation of log output from the log of natural output level.

Proof. We start with the optimal price setting equation (11)

$$x_{it} = (1-\beta\theta)\sum_{\tau=t}^{\infty}(\beta\theta)^{\tau-t}\mathbb{E}_t p_\tau^*$$

$$
\begin{aligned}
x_{it} - p_{t-1} &= (1-\beta\theta)\sum_{\tau=t}^{\infty}(\beta\theta)^{\tau-t}\mathbb{E}_t(p^*_\tau - p_{t-1}) \\
&= (1-\beta\theta)(p^*_t - p_{t-1}) + \beta\theta\left[(1-\beta\theta)\sum_{\tau=t+1}^{\infty}(\beta\theta)^{\tau-(t+1)}\mathbb{E}_t(p^*_\tau - p_t)\right] \\
&+ \beta\theta(1-\beta\theta)\sum_{\tau=t+1}^{\infty}(\beta\theta)^{\tau-t-1}\mathbb{E}_t(p_t - p_{t-1}) \\
&= (1-\beta\theta)(p^*_t - p_{t-1}) + \beta\theta\mathbb{E}_t(x_{i,t+1} - p_t) + \beta\theta(p_t - p_{t-1}) \\
\Rightarrow \frac{\pi_t}{1-\theta} &= (1-\beta\theta)(p^*_t - p_{t-1}) + \beta\theta\mathbb{E}_t\frac{\pi_{t+1}}{1-\theta} + \beta\theta\pi_t \\
\Rightarrow \pi_t &= \frac{(1-\beta\theta)(1-\theta)}{1-\beta\theta(1-\theta)}(p^*_t - p_{t-1}) + \frac{\beta\theta}{1-\beta\theta(1-\theta)}\mathbb{E}_t\pi_{t+1} \\
\Rightarrow \pi_t &= \frac{(1-\beta\theta)(1-\theta)}{1-\beta\theta(1-\theta)}(p^*_t - p_t + \underbrace{p_t - p_{t-1}}_{=\pi_t}) + \frac{\beta\theta}{1-\beta\theta(1-\theta)}\mathbb{E}_t\pi_{t+1} \\
\Rightarrow \pi_t &= \frac{(1-\beta\theta)(1-\theta)}{\theta}(p^*_t - p_t) + \beta\mathbb{E}_t\pi_{t+1}
\end{aligned}
$$

Next, we have to show that $p^*_t - p_t = (\sigma+\phi)(y_t - y^n_t)$, y^n_t is the log of natural output level under flexible pricing.

From the first order conditions of the households we have

$$
\begin{aligned}
\frac{W_t}{P_t} &= \psi C_t^\sigma N_t^\phi \\
\Rightarrow \frac{W_t}{P^*_t}\frac{P^*_t}{P_t} &= \psi C_t^\sigma N_t^\phi \\
\Rightarrow \frac{P^*_t}{P_t} &= \frac{\varepsilon\psi}{\varepsilon-1}\frac{C_t^\sigma N_t^\phi}{A_t}
\end{aligned}
$$

Using Claim 1 and goods market clearing condition, we have,

$$
\Rightarrow \frac{P^*_t}{P_t} = \frac{\varepsilon\psi}{\varepsilon-1}\frac{Y_t^\sigma Y_t^\phi}{A_t^{1+\phi}}.
$$

Under flexible price equilibrium, $P^*_t = P_t$, which results in $Y_t^{n(\sigma+\phi)} = \dfrac{\varepsilon-1}{\varepsilon\psi}A_t^{1+\phi}$.

Hence, we get

$$
\frac{P^*_t}{P_t} = \frac{Y_t^{\sigma+\phi}}{Y_t^{n(\sigma+\phi)}}.
$$

Taking log and substituting it back into the expression for π_t, we get the NK Phillips

Curve

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \pi_{t+1},$$

where $\kappa = \dfrac{(1-\theta)(1-\beta\theta)(\sigma+\phi)}{\theta}$. ■

Dynamic IS relation

Taking log of (6), we get the IS relation

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - \rho), \tag{13}$$

where $i_t = -\log Q_t$, and $\rho = -\log \beta$. Applying market clearing condition we then get

$$y_t = \mathbb{E}_t y_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - \rho), \tag{14}$$

The dynamic IS relation expressed in terms of output gap becomes

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\sigma}(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n), \tag{15}$$

where r_t^n is the natural real rate of interest given by

$$r_t^n = \rho + \frac{1+\phi}{\sigma+\phi}\mathbb{E}_t(a_{t+1} - a_t). \tag{16}$$

Interest rate rule

It is assumed that the central bank follows an interest rate rule of following form

$$i_t = \rho + \phi_\pi \pi_t + \phi_y \tilde{y}_t + \nu_t, \tag{17}$$

where ν_t is an AR(1) process with zero unconditional mean. In addition, it is also assumed that ϕ_π, and ϕ_y are non negative constants chosen by the central bank.

$$\nu_t = \rho_\nu \nu_{t-1} + \epsilon_t^\nu. \tag{18}$$

Equilibrium dynamics

Combining equation (17), (12), and (15) we get the following linear rational expectation model

$$\begin{bmatrix} \nu_{t+1} \\ \mathbb{E}_t \pi_{t+1} \\ \mathbb{E}_t \tilde{y}_{t+1} \end{bmatrix} = \begin{bmatrix} \rho_\nu & 0 & 0 \\ 0 & \frac{1}{\beta} & -\frac{\kappa}{\beta} \\ 1 & \frac{1}{\sigma}\left(\phi_\pi - \frac{1}{\beta}\right) & 1 + \frac{1}{\sigma}\left(\phi_y - \frac{\kappa}{\beta}\right) \end{bmatrix} \begin{bmatrix} \nu_t \\ \pi_t \\ \tilde{y}_t \end{bmatrix} + \begin{bmatrix} \epsilon_{t+1}^\nu \\ 0 \\ 0 \end{bmatrix}; \tag{19}$$

with $\nu_0 = \epsilon_t^\nu$. Where $A_t = 1 \; \forall t$ has been assumed for simplicity.

Results

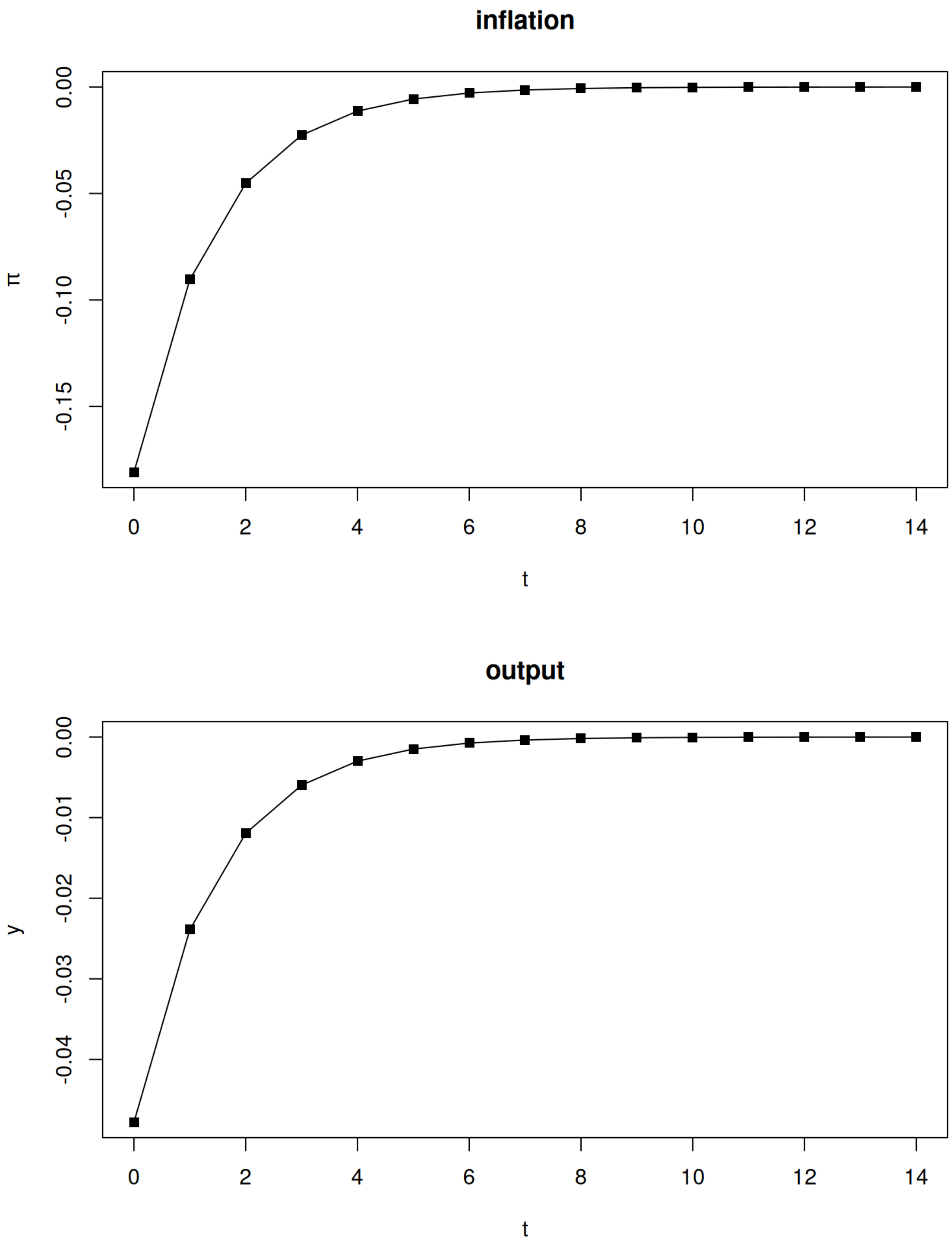

Figure 1: Contractionary interest rate shock

3 A model with inventory

The description of households and final good firm is identical to the description in Section 2. However now we have two types of intermediate good firms - manufacturing and services. All firms use identical production technology which is linear in labor input. Manufacturing firms carry out their production activity in remote location and demand for manufactured good is fulfilled through retail units. On the other hand, the service firms produce their output at the location of demand.

Household

There are a continuum of of mass 1 household that maximizes their lifetime utility over consumption and leisure. The expected lifetime utility is given by:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \psi \frac{N_t^{1+\phi}}{1+\phi} \right), \tag{20}$$

where β is the discount factor, $\sigma > 0$ is that elasticity of intertemporal substitution, and $\phi > 0$ is the Frisch elasticity of labor supply. The household's optimization problem is subject to the budget constraint given by:

$$P_t C_t + Q_t B_t = W_t N_t + B_{t-1} + P_t \mathcal{F}_t \tag{21}$$

where B_t is the number of shares of government bond, Q_t is the price of government bond, and $\mathcal{F}_t$ is the profits of the firm.

Let $\Pi_t = \dfrac{P_t}{P_{t-1}}$ denote the inflation rate. The households' choice can be characterized by the following first order conditions

$$-\frac{U_{n,t}}{U_{c,t}} = \frac{W_t}{P_t} = \psi C_t^{\sigma} N_t^{\phi}, \tag{22}$$

$$Q_t = \beta \mathbb{E}_t \left(\frac{U_{c,t+1}}{U_{c,t}} \frac{P_t}{P_{t+1}} \right). \tag{23}$$

Let the household allocate N_{mt} units of labor to the manufacturing sector, and N_{st} units of labor to the service sector, so that $N_{mt} + N_{st} = N_t$.

The final good producer

The final good (Y_t) is produced using the intermediate goods Y_{it} using the following technology:

$$Y_t = \left(\int_0^1 Y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{24}$$

where ε is the elasticity of substitution.

Final good producers are perfectly competitive in the input and output market. They maximize profits subject to the production function (24). Therefore the input demand function is given by:

$$Y_{it} = \left(\frac{P_{it}}{P_t} \right)^{-\varepsilon} Y_t \quad \forall i, \tag{25}$$

and the price index is

$$P_t = \left(\int_0^1 P_{it}^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}}. \tag{26}$$

Since the intermediate output comes from two sectors, the final good output can also be expressed as follows

$$Y_t = \left(\underbrace{\int_0^{\vartheta} Y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di}_{\text{manufaturing output}} + \underbrace{\int_{\vartheta}^1 Y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di}_{\text{service output}} \right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{27}$$

where ϑ is the fraction of manufacturing firms[1].

Since the output of individual sectors differ only in their index, therefore the quantity and price of each product within a sector will be same. That is the demand for manufacturing output from the firm will be

$$Y_{imt} = \left(\frac{P_{mt}}{P_t} \right)^{-\varepsilon} Y_t \quad \forall i, \tag{28}$$

[1]The notion of final good output used here is different from GDP which will be $Y_t = \left(\underbrace{\int_0^{\vartheta} Y_{i,t+1}^{\frac{\varepsilon-1}{\varepsilon}} di}_{\text{intermediate output}} + \underbrace{\int_{\vartheta}^1 Y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di}_{\text{service output}} \right)^{\frac{\varepsilon}{\varepsilon-1}} = \left(\underbrace{\int_0^{\vartheta} \xi_{it}^{\frac{\varepsilon-1}{\varepsilon}} di}_{\text{intermediate output}} + \underbrace{\int_{\vartheta}^1 Y_{it}^{\frac{\varepsilon-1}{\varepsilon}} di}_{\text{service output}} \right)^{\frac{\varepsilon}{\varepsilon-1}}$

and the demand for service output of each intermediate good firm from the service sector will be

$$Y_{ist} = \left(\frac{P_{st}}{P_t}\right)^{-\varepsilon} Y_t \quad \forall i. \tag{29}$$

Finally the price index can be decomposed as

$$P_t = \left(\vartheta P_{mt}^{1-\varepsilon} + (1-\vartheta) P_{st}^{1-\varepsilon}\right)^{\frac{1}{1-\varepsilon}}. \tag{30}$$

Intermediate good firms

There are two types of intermediate goods - manufactured goods and service goods. Total mass of the intermediate good firms is 1, a fraction ϑ of them are manufacturing and rest are service firms. The manufacturing activity takes place at a location that is different from the location of demand. So these firms must rely on retail units to meet the demand from the final consumers. The retail units face the consumer demand and orders inventory at discrete intervals of time once the stock becomes too low. For each order it has to pay administrative cost, independent of the quantity ordered, on top of the per unit price. The entire payment is made at the beginning of the next period [2]. And the ordered inventory is delivered at the end of the period. The stock of inventory thus procured will be available for final consumption in the next period.

Manufacturing unit

The manufacturing produces a single good as output using labor as only input. The production technology is constant returns to scale

$$X_{imt} = A_t N_{imt}.$$

The production process takes place at a distant location away from the households. Hence retail units are required for goods to be delivered to the final consumers. The manufacturing unit has to fulfill the quantity demanded by the retail units at discrete interval of time. Suppose at time t_i the retail unit orders ξ_i quantity of goods then the manufacturing firm's optimization problem

[2] This assumption is needed to simplify the characterization of retailer's problem.

can be stated as follows:

$$\pi_{im} = \max_{X_{imt}, N_{imt}, c_{t_i}} \left\{ \sum_{t=0}^{\infty} \beta^t (-W_t N_{imt}) + \sum_{i=1}^{\infty} \beta^{t_i} c_{t_i} \xi_i + k \right\} \tag{31}$$

$$\text{subject to} \quad X_{imt} = A_t N_{imt}.$$

where W_t is the labor wage rate, N_{imt} is the labor employed by unit i in period t, and c_{t_i} is the per unit cost of goods at the manufacturer's location. Marker clearing implies

$$\xi_i = \sum_{t=t_{i-1}}^{t_i} X_{imt}; \quad \forall i = 0, 1, \dots,$$

and $t_0 = 0$.

Assumption 1. *Manufacturing unit behaves competitively in the input (labor) market.*

Claim 3. *Under Assumption 1, the solution to the manufacturing unit's optimization problem is characterized by the following equation:*

$$\beta^{t_i} c_{t_i} A_t = \beta^t W_t.$$

Because each manufacturing firm is identical, therefore the labor supply to manufacturing firm will be $N_{imt} = \dfrac{1}{\vartheta} N_{mt}$.

Retail unit

The retail unit begins with a positive stock of inventory, x_0 and faces the consumer demand. If the stock size is too low, then it places an order with the manufacturing unit at the beginning of the period which costs k irrespective of the size of the order, it addition it also pays a per unit cost c_t. Once the order is fulfilled, the retailer also incurs a storage, $f(x) = \gamma x$, where x is the existing stock of inventory. The retailer's optimization problem can be states as follows:

$$\pi_r(x_{im0}) = \max_{P_{imt}, \{t_i, \xi_i\}_{i=1}^{\infty}} \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t [-P_{imt}\Delta x_{imt} - f(x_t + \Delta x_{imt})] - \sum_{i=1}^{\infty} \beta^{t_i}(k + c_{t_i}\xi_i) \right\} \tag{32}$$

subject to

$$\Delta x_{imt} = -\left(\frac{P_{imt}}{P_t}\right)^{-\varepsilon} Y_t,$$

where $\varepsilon > 1$.

The solution to the retailer's problem is determined as follows. Consider the interval $t_i < t \leq t_{i+1}$, the retailer orders the inventory at the beginning of t_i, the goods delivered will be available in $t_i + 1$ and sold by the time t_{i+1}. And a new inventory will be ordered at the beginning of t_{i+1}. The firm's reward in this interval will be:

$$\sum_{t=t_i+1}^{t_{i+1}} \beta^{(t-t_i-1)} \mathbb{E}_{t_i}[-P_{imt}\Delta x_{imt} - f(x_t + \Delta x_{imt})] - (c_{t_i}\xi_i - k) \tag{33}$$

with $x_{t_i+1} = \xi_i$.

$$= \sum_{t=t_i+1}^{t_{i+1}} \beta^{(t-t_i-1)} \mathbb{E}_{t_i}[\underbrace{-P_{imt}\Delta x_{imt}}_{Revenue} - \underbrace{f(x_t + \Delta x_{imt})}_{\text{Storage cost}}] - \underbrace{\sum_{t=t_{i-1}+1}^{t_i} c_{t_i} X_{imt} - k}_{\text{Procurement cost}}$$

From the above equation, observe that it is optimal for the retailer to set its price such that the entire stock of output is sold out in the current period, so that the revenue is maximized and the storage cost is minimized.

Claim 4. *The optimal strategy of the retailer is to set the price P_{imt} such that the entire inventory is sold out in the current period.*

Proof. Given that the rate of the demand is proportional to $P_{imt}^{-\varepsilon}$, the revenue in any period is then, proportion to $P_{imt}^{1-\varepsilon}$ which is a decreasing function of the price, P_{imt}.

Secondly, the storage cost, $f(x)$, is an increasing function of the amount of inventory being stored.

Lastly, the cost of accumulating inventory in period t_i, $\sum_{t=t_{i-1}+1}^{t_i} c_{t_i} X_{imt} - k$ has already been incurred at the beginning of period $t_i + 1$, and therefore it acts as a fixed quantity in the interval under consideration.

Thus the retailer's optimization will imply that it sets price low enough to sell all the inventory in the current period, which simultaneously maximizes the revenue and reduces the storage cost. ■

Hence, we conclude that, in every period t the retail unit orders a new inventory ξ_{imt} from the manufacturing unit and transforms the existing inventory $\xi_{im,t-1}$ into the final manufactured good $Y_{imt} = \xi_{im,t-1}$. It can be shown that for the linear production function assumed in this section the profit maximizing price, based on expected demand of manufactured goods, can be written as

$$P^{\star}_{m,t+1} = \frac{\varepsilon}{\varepsilon(1-\gamma)-1} \frac{W_t}{A_t}. \tag{34}$$

where γ is the per unit storage cost of inventory. And the actual price based on realized demand of manufactured goods will be

$$P_{mt} = \left(\frac{Y_{imt}}{Y_t}\right)^{-\frac{1}{\varepsilon}} P_t = \left(\frac{X_{t-1}}{Y_t}\right)^{-\frac{1}{\varepsilon}} P_t. \tag{35}$$

The expected demand function for the manufactured goods is

$$Y_{mt+1} = \mathbb{E}\left[\left(\frac{P^{\star}_{m,t+1}}{P_{t+1}}\right)^{-\varepsilon} Y_{t+1}\right].$$

Therefore, the quantity of inventory order, based on future expectation is

$$\xi_{imt} = \frac{\varepsilon}{\varepsilon(1-\gamma)-1} \mathbb{E}\left[\left(\frac{W_t}{A_t P_{t+1}}\right)^{-\varepsilon} Y_{t+1}\right]. \tag{36}$$

Claim 5. *The inventory ordered at the beginning of period t is*

$$\xi_{imt} = \frac{\varepsilon}{\varepsilon(1-\gamma)-1}\mathbb{E}\left[\left(\frac{W_t}{A_t P_{t+1}}\right)^{-\varepsilon} Y_{t+1}\right].$$

This inventory will be delivered to the retailer by the end of the period t. The expenses incurred will be paid at the beginning of period $t+1$.

Service sector

In this paper service sector does not maintain an inventory and therefore supply and demand should match every period. The absence of inventory then implies that the price of service sector goods will be

$$P_{st} = \frac{\varepsilon}{\varepsilon - 1}\frac{W_t}{A_t}. \tag{37}$$

Equilibrium

Definition 1. *A competitive equilibrium for the given setup with inventory is the sequence of price $\mathbb{P} = \{P_t, W_t, Q_t, P_{mt}, P_{st}, c_t\}$; household choices $\mathbb{Z}_h(B_{t-1}) = \{C_t, N_t, B_t\}$; manufacturing unit choice $\mathbb{Z}_m = \{X_{imt}, N_{imt}\}$; retail unit choices $\mathbb{Z}_r(\xi_{im,t-1}) = \{P_{imt}, \xi_{imt}, Y_{imt}\}$; and service sector firm choices $\mathbb{Z}_s = \{Y_{ist}, N_{ist}\}$, such that*

- *Given $\mathbb{P}$, households maximize their utility*
- *Given $\mathbb{P}$, manufacturing unit gets zero profit*
- *Given $\mathbb{P}$, retail units and service sector firms maximize their profits*
- *Prices $\mathbb{P}$ are such that markets clears*

    1. *Manufactured goods market: $Y_{imt} = C_{imt}$.*
    2. *Service goods market: $Y_{ist} = C_{ist}$.*
    3. *Labor market: $N_t = N_{mt} + N_{st}$.*
    4. *Inventory market: $\xi_{imt} = X_{imt}$.*

The price of bonds Q_t is determined by government policy.

Proposition 1. *The manufacturing price is sticky*

$$P_{mt} = \left(\frac{\varepsilon\psi}{\varepsilon(1-\gamma)-1} \frac{Y_{t-1}^{\sigma} N_{t-1}^{\phi}}{A_{t-1}} \right) P_{t-1}$$

The service sector price is not sticky

$$P_{st} = \left(\frac{\varepsilon\psi}{\varepsilon-1} \frac{Y_t^{\sigma} N_t^{\phi}}{A_t} \right) P_t.$$

The aggregate price is sticky

$$P_t^{1-\varepsilon} = \vartheta P_{mt}^{1-\varepsilon} + (1-\vartheta) P_{st}^{1-\varepsilon}$$
$$\left(1 - (1-\vartheta) \left(\frac{\varepsilon\psi}{\varepsilon-1} \frac{Y_t^{\sigma} N_t^{\phi}}{A_t} \right)^{1-\varepsilon} \right) P_t^{1-\varepsilon} = \vartheta \left(\frac{\varepsilon\psi}{\varepsilon(1-\gamma)-1} \frac{Y_{t-1}^{\sigma} N_{t-1}^{\phi}}{A_{t-1}} \right)^{1-\varepsilon} P_{t-1}^{1-\varepsilon}.$$

Phillips curve

In order to derive the Phillips curve, the first order approximation around zero inflation steady state is taken.

Zero inflation steady state

In this section the expressions from variables like output, employment, and price index are derived in zero inflation steady state.

Definition 2. *A zero inflation steady state is the economy in which the prices of each sector remains constant.*

The price of each sectoral good are given by

$$P_m = \frac{\varepsilon}{\varepsilon(1-\gamma)-1} \frac{W}{A}; \tag{38}$$
$$P_s = \frac{\varepsilon}{\varepsilon-1} \frac{W}{A}. \tag{39}$$

From here, the price index may be written as

$$P^{1-\varepsilon} = \vartheta P_m^{1-\varepsilon} + (1-\vartheta) P_s^{1-\varepsilon}. \tag{40}$$

Given the expression for prices we can now derive the demand functions for manufactured and service sector goods.

$$Y_m = \left(\frac{P_m}{P}\right)^{-\varepsilon} Y \tag{41}$$

$$Y_s = \left(\frac{P_s}{P}\right)^{-\varepsilon} Y. \tag{42}$$

And the employment in each sector would be

$$N_m = \vartheta \left(\frac{P_m}{P}\right)^{-\varepsilon} \frac{Y}{A} \tag{43}$$

$$N_s = (1-\vartheta) \left(\frac{P_s}{P}\right)^{-\varepsilon} \frac{Y}{A}. \tag{44}$$

Definition 3. *The natural rate of output is defined as the level of output at zero inflation price level.*

Lemma 1. *Upto first order approximation*

$$p_{mt} = p_{t-1} + (1-\sigma)\tilde{y}_t. \tag{45}$$

$$p_{st} = p_t + (\sigma + \phi)\tilde{y}_t. \tag{46}$$

Proof.

$$
\begin{aligned}
P_{mt} &= \frac{\varepsilon}{\varepsilon(1-\gamma)-1} \frac{W_{t-1}}{A_{t-1}P_{t-1}} P_{t-1} \\
&= \frac{\varepsilon\psi}{\varepsilon(1-\gamma)-1} \frac{Y_{t-1}^{\sigma} N_{t-1}^{\phi}}{A_{t-1}} P_{t-1} \\
P_m e^{p_{mt}} &= \frac{\varepsilon\psi}{\varepsilon(1-\gamma)-1} \frac{Y^{\sigma} N^{\phi}}{A} P e^{(\sigma y_{t-1} + \phi n_{t-1} - a_{t-1} + p_{t-1})} \\
\Rightarrow p_{mt} &= p_{t-1} + \sigma y_{t-1} + \phi n_{t-1} - a_{t-1}
\end{aligned}
$$

Define $y_t^n = a_t - \phi n_t$ as natural level of final output around steady state.

$$
\begin{aligned}
\Rightarrow p_{mt} &= p_{t-1} + \sigma \tilde{y}_{t-1}; \quad \tilde{y}_t = y_t - y_t^n \\
\text{Similarly,} P_{st} &= \frac{\varepsilon}{\varepsilon - 1} \frac{W_t}{A_t} \\
\Rightarrow p_{st} &= p_t + \sigma y_t + \phi n_t - a_t
\end{aligned}
$$

■

The Phillips curve can be derived as follows

Claim 6. *The Phillips curve for the model with inventory is given by*

$$
\mathbb{E} y_{t+1} + \mathbb{E} \varepsilon \pi_{t+1} - x_t = \varepsilon(\sigma y_t + \phi n_t - a_t)
$$

$$
\Rightarrow \mathbb{E} \pi_{t+1} = \tilde{y}_t - \frac{1}{\varepsilon}(\mathbb{E} y_{t+1} - x_t),
$$

where $\tilde{y}_t$ is the consumption gap.

Proof. See Appendix A. ■

Results and Discussion

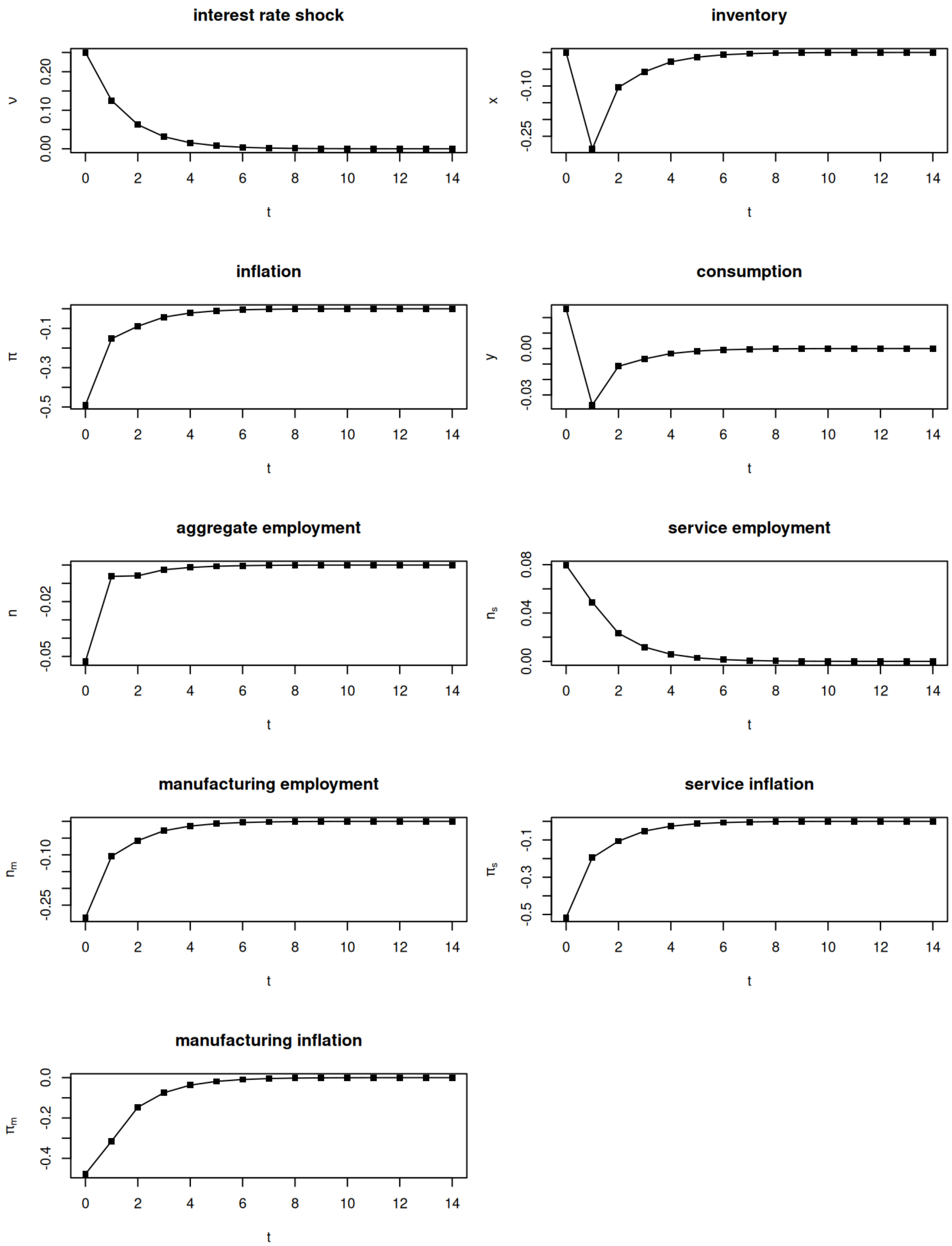

Figure 2: Contractionary interest rate shock.

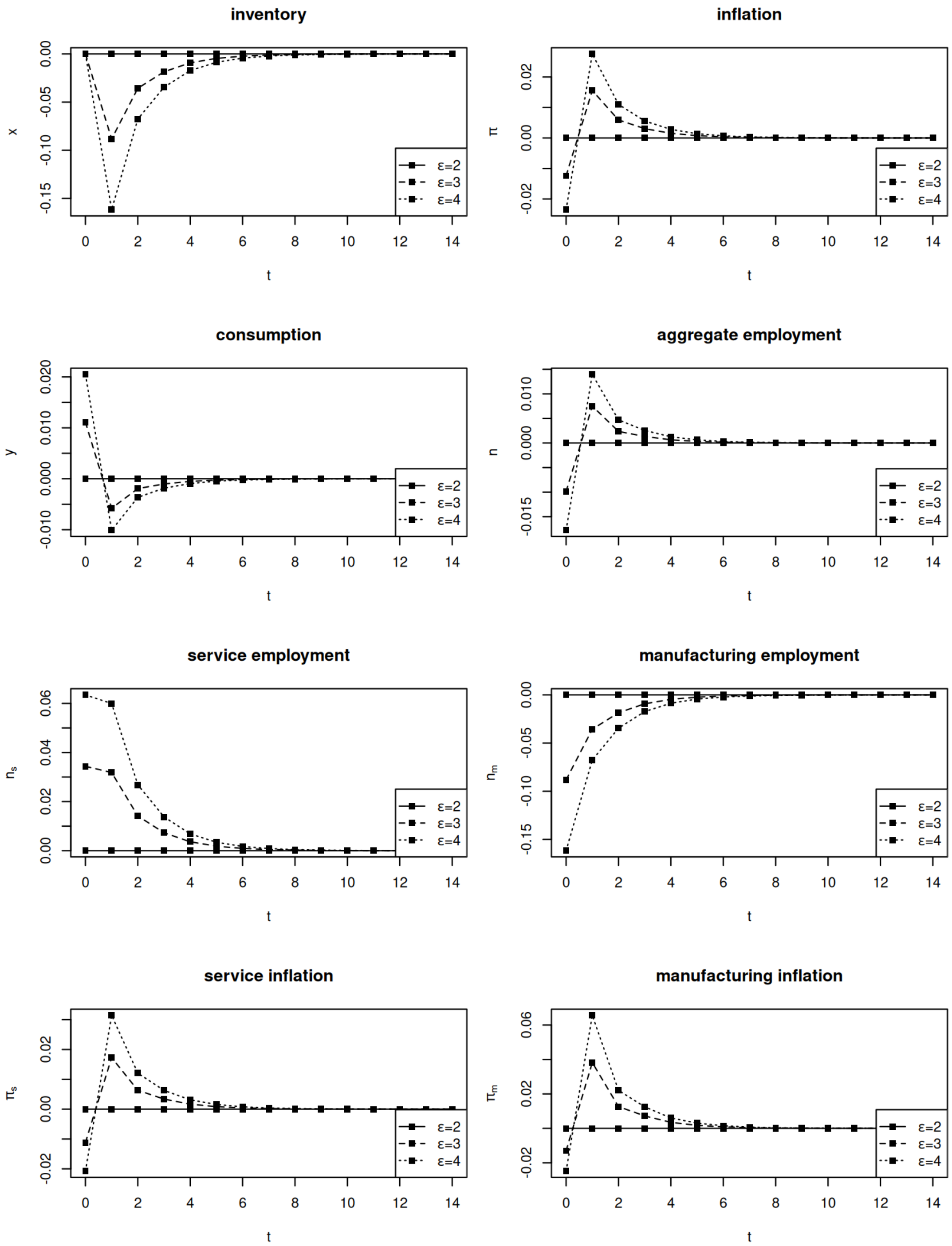

Figure 3: Change in elasticity of substitution.

From the Figure 2, we can see than in response to contractionary monetary policy shock inflation and aggregate employment reduces as expected. The manufactured goods prices are sticky and therefore the manufacturing firms have limited ability to reduce the prices in comparison to price reduction in service sector. The response of manufacturing inflation is weaker than the response of service inflation. Recall that a contractionary monetary policy shock induces consumers to reduce there aggregate consumption expenditure. Since there is limited scope of price reduction in manufacturing sector, therefore the consumption and hence the output (and employment) in manufacturing sector must decline. Whereas the service sector is able to reduce there prices sharply, hence it is possible to increase service output (and employment) in such a manner that the service firm profit goes up. Is that a contradiction? No, even though the consumption of services goes up, the aggregate consumption expenditure which is a function of prices and quantity goes down. This is as expected from the fundamental principle of substitution effect. This is something we can also see from the plots. At $t = 0$, when the shock hits the aggregate consumption goes up by about 3% but the aggregate price goes down by 50%, so the aggregate consumption expenditure goes down by 47%.

The initial response of aggregate consumption is driven by increase in service output. The manufacturing consumption is fixed at the level of inventory held by the retailers at $t = 0$, so an increase in service output increases overall consumption. However, from $t = 1$ onward the manufactured output (and inventory) declines by about 30% but the service output is up by about 5% only, therefore the aggregate consumption goes down.

Figure 3 shows the output for different values of elasticity of substitution (ε) relative to $\varepsilon = 2$. That is we subtract the $\varepsilon = 2$ reponse from the response to policy shock when $\varepsilon = 3, 4$. This is done for clarity of plots and ease of interpretation. The observation from here is that the initial response gets stronger as the elasticity increases[3].

The initial manufactured good consumption is fixed at the level of inventory carried forward that cannot be changed, but as ε, the manufactured goods and services tends to become substitute for each other, hence in order to insure that entire inventory is sold out the retailers responds with higher reduction in price. This induces the service sector to reduce their prices sharply as well at $t = 0$.

From $t = 1$, onward the inventory accumulation is reduced, therefore the prices stabilizes faster

[3] As ε increases from 0 to 1, we move from the case of perfect complement to Cobb Douglas aggregation function. Furthermore, as ε moves to 1 to ∞, the manufactured goods and services become perfect substitutes for the consumer.

as elasticity increases.

4 Conclusion

In this paper, we have constructed a nominal model with manufacturing and service sector. We first show that prices are endogenously sticky. Then using the impulse response functions, we find that the response of manufacturing sector in terms of employment is stronger than the response of service sector. However service inflation responds more strongly than the manufacturing inflation, as prices in service sector are more flexible. Thirdly, we find that as the elasticity of substitution increases, the initial response becomes stronger due to fixed value of initial inventory/ manufacturing consumption. But the variables returns to their steady state faster.

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A Characterization and Log linearization

The key equations characterizing the inventory model are:

$$\xi_t = \left(\frac{\varepsilon}{\varepsilon(1-\gamma)-1} \frac{W_t}{A_t P_{t+1}} \right)^{-\varepsilon} Y_{t+1}, \tag{47}$$

$$Q_t = \beta \mathbb{E}_t \left[\frac{Y_t^{\sigma}}{Y_{t+1}^{\sigma}} \frac{P_t}{P_{t+1}} \right], \tag{48}$$

$$\frac{W_t}{P_t} = \psi Y_t^{\sigma} N_t^{\phi}, \tag{49}$$

$$P_{mt}^{-\varepsilon} = \frac{\xi_{t-1}}{Y_t} P_t^{-\varepsilon}, \tag{50}$$

$$P_{st}^{-\varepsilon} = \frac{A_t N_{st}}{Y_t} P_t^{-\varepsilon}, \tag{51}$$

$$P_{st} = \frac{\varepsilon}{\varepsilon - 1} \frac{W_t}{A_t}, \tag{52}$$

$$N_t = \frac{\xi_t}{A_t} + N_{st}, \tag{53}$$

$$Y_t^{\frac{\varepsilon-1}{\varepsilon}} = \vartheta \xi_{t-1}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\vartheta) Y_{st}^{\frac{\varepsilon-1}{\varepsilon}}. \tag{54}$$

Variables: Y_t, N_t, N_{st}, ξ_t

P_t, W_t, P_{mt}, P_{st}.

Log linearization

$$
\begin{aligned}
x_t &= y_{t+1} - \varepsilon(w_t - a_t - p_{t+1}), \\
y_t &= y_{t+1} - \frac{1}{\sigma}\left[i_t - \pi_{t+1} - \rho\right], \\
w_t - p_t &= \sigma y_t + \phi n_t, \\
p_{mt} - p_t &= \frac{1}{\varepsilon}(y_t - x_{t-1}), \\
p_{st} - p_t &= \frac{1}{\varepsilon}(y_t - a_t - n_{st}), \\
p_{st} &= w_t - a_t, \\
N n_t &= N_m(x_t - a_t) + N_s n_{st}, \\
Y y_t &= \vartheta A N_m x_{t-1} + (1-\vartheta) A N_s (a_t + n_{st}),
\end{aligned}
$$

Variables: y_t, n_t, n_{st}, x_t

p_{mt}, p_{st}, p_t, w_t.

Eliminating p_{mt}, p_{st}. The above set of equations may be written as

$$
\begin{aligned}
y_{t+1} + \varepsilon p_{t+1} - x_t &= \varepsilon(w_t - a_t), \\
y_{t+1} + \frac{1}{\sigma} p_{t+1} &= y_t + \frac{1}{\sigma}[i_t + p_t - \rho], \\
0 &= \sigma y_t + \phi n_t + p_t - w_t, \\
0 &= \frac{1}{\varepsilon}(y_t - a_t - n_{st}) + p_t - w_t + a_t, \\
0 &= N_m(x_t - a_t) + N_s n_{st} - N n_t, \\
0 &= -Y y_t + \vartheta A N_m x_{t-1} + (1-\vartheta) A N_s (a_t + n_{s,t}).
\end{aligned}
$$

with Taylor's rule: $i_t = \rho + \phi_\pi \pi_t + \phi_y y_t$.

Eliminating $p_t - w_t$

$$\begin{aligned} y_{t+1} + \varepsilon\pi_{t+1} - x_t &= \varepsilon(\sigma y_t + \phi n_t - a_t), \\ y_{t+1} + \frac{1}{\sigma}\pi_{t+1} &= y_t + \frac{1}{\sigma}[i_t - \rho], \\ 0 &= \left(\frac{1}{\varepsilon} - \sigma\right) y_t + \left(1 - \frac{1}{\varepsilon}\right) a_t - \frac{1}{\varepsilon} n_{st} - \phi n_t, \\ -N_m x_t &= -N_m a_t + N_s n_{st} - N n_t, \\ 0 &= -Y y_t + \vartheta A N_m x_{t-1} + (1-\vartheta) A N_s (a_t + n_{s,t}). \end{aligned}$$

with Taylor's rule: $i_t = \rho + \phi_\pi \pi_t + \phi_y y_t$.

Define $z_t = \begin{bmatrix} a_t & \nu_t & x_{t-1} & \pi_t & y_t & n_t & n_{st} \end{bmatrix}'$ and $z_{2t} = \begin{bmatrix} n_{mt} & \pi_{mt} & \pi_{st} \end{bmatrix}'$. The above system of equations can be written as

$$G z_{t+1} = H z_t$$

where

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & \varepsilon & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\sigma} & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -N_m & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} \rho_a & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \rho_\nu & 0 & 0 & 0 & 0 & 0 \\ -\varepsilon & 0 & 0 & 0 & \sigma\varepsilon & \phi\varepsilon & 0 \\ 0 & \frac{1}{\sigma} & 0 & \frac{\phi_\pi}{\sigma} & 1 + \frac{\phi_y}{\sigma} & 0 & 0 \\ 1 - \frac{1}{\varepsilon} & 0 & 0 & 0 & \frac{1}{\varepsilon} - \sigma & -\phi & -\frac{1}{\varepsilon} \\ -N_m & 0 & 0 & 0 & 0 & -N & N_s \\ (1-\vartheta)AN_s & 0 & \vartheta AN_m & 0 & -Y & 0 & (1-\vartheta)AN_s \end{bmatrix}$$